Grade 10 – Term 1: C1–C2 Classroom Activity – Version 2

(C1) *Identify the decision alternatives, the states of nature, the uncertain event, and the payoff components.*

(C2) *construct the corresponding payoff table.*

Context Descriptions

All per-unit contributions and adjustments are already net of relevant per-unit costs. Fixed amounts apply once over the stated activity period. All state adjustments apply to every alternative. No other monetary components are required.

1. A textbook publisher must choose a digital licence or a print licence before the education board announces whether it will retain or revise its syllabus. These are external syllabus decisions. Analysts have already calculated the complete profit for each outcome, in thousand USD. The digital licence yields 23 thousand USD if the syllabus is retained and 14 thousand USD if it is revised. The print licence yields 34 thousand USD if the syllabus is retained and 9 thousand USD if it is revised.
2. A rehearsal studio must choose a daytime contract or an evening contract before an independent organiser announces an extended or shortened festival. The daytime contract provides a fixed net contribution of 640 thousand JPY; the evening contract provides a fixed net contribution of 480 thousand JPY. An extended festival adds a fixed access payment of 210 thousand JPY to either contract. A shortened festival instead imposes a fixed cancellation deduction of 120 thousand JPY on either contract. No output quantity enters these fixed payments.
3. An archive contractor must select a flatbed scanner or an overhead camera before ambient humidity during a project is known. The external conditions will be dry air or humid air. The flatbed scanner contributes 7 thousand CNY per completed archive collection; the overhead camera contributes 9 thousand CNY per completed collection. With the flatbed scanner, 13 collections would be completed in dry air and 9 in humid air. With the overhead camera, 11 would be completed in dry air and 7 in humid air. Dry air triggers a fixed preservation-contract bonus of 8 thousand CNY; humid air causes a fixed climate-control loss of 7 thousand CNY. Each fixed adjustment applies to either equipment choice.
4. A ceramics manufacturer must choose a tableware line or a tile line before external wholesale orders are known. Orders will be brisk or quiet. The tableware line contributes 5 thousand EUR per completed firing batch and a fixed partner contribution of 9 thousand EUR. The tile line contributes 7 thousand EUR per completed firing batch and a fixed partner contribution of 13 thousand EUR. Brisk orders add 4 thousand EUR per batch to either line; quiet orders add 2 thousand EUR per batch to either line. The tableware line would complete 18 batches with brisk orders and 11 with quiet orders. The tile line would complete 15 and 8 batches under those respective conditions.
5. A vocational training centre must choose a woodworking programme or a metalworking programme before the regional authority announces increased, baseline, or reduced training funding. The woodworking programme contributes 6 thousand GBP per completed course and a fixed partner payment of 9 thousand GBP. The metalworking programme contributes 7 thousand GBP per completed course and a fixed partner payment of 11 thousand GBP. For either programme, increased funding adds 3 thousand GBP per course and a fixed administration grant of 8 thousand GBP. Baseline funding adds 2 thousand GBP per course and has no fixed state adjustment. Reduced funding deducts 2 thousand GBP per course and imposes a fixed administration loss of 6 thousand GBP. Under increased, baseline, and reduced funding respectively, the centre would complete 12, 8, and 3 woodworking courses, or 16, 11, and 6 metalworking courses. These are the complete conditional course quantities.

6. A road resurfacing contractor must commit to a portable mixer, a towable mixer, or a batching plant before the weather during the work period is known. External conditions will be dry weather (favorable), mixed weather (neutral), or persistent rain (adverse). The portable mixer contributes 7 million COP per completed road section and requires a fixed commitment of 25 million COP. The towable mixer contributes 8 million COP per section and requires a fixed commitment of 38 million COP. The batching plant contributes 9 million COP per section and requires a fixed commitment of 52 million COP. For every equipment choice, dry weather adds 3 million COP per section and a fixed completion bonus of 11 million COP; mixed weather has no per-section or fixed adjustment; persistent rain deducts 3 million COP per section and causes a fixed disruption loss of 14 million COP. Under dry weather, mixed weather, and persistent rain respectively, the portable mixer would complete 16, 11, and 4 sections; the towable mixer would complete 20, 14, and 5 sections; and the batching plant would complete 24, 19, and 6 sections. Deduct each equipment commitment once, in every state.

1 Solutions: Extraction – Version 2

Problems:

1.1. Textbook publishing rights (C1–C2)

1.2. Rehearsal studio contract (C1–C2)

1.3. Archive imaging equipment (C1–C2)

1.4. Ceramics production line (C1–C2)

1.5. Vocational training programme (C1–C2)

1.6. Road resurfacing equipment (C1–C2)

1.1 Exercise 1: Textbook publishing rights

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Digital licence	A_2: Print licence
States of nature	S_1: Syllabus retained	S_2: Syllabus revised
Uncertain event	Whether the education board retains or revises its syllabus after the publishing licence is chosen	
Consequences	Profit in thousand USD for each alternative–state pair. All four profits are supplied directly.	

C2 - Building the payoff table.

Relevant values. The context supplies the complete profit for every licence–syllabus pair.

Abstract component	Context component	Math symbol	Corresponding value
Direct payoff	Complete profit for the chosen licence under the board's syllabus decision	$P_{ay}(A_i, S_j)$	Digital–retained: 23 thousand USD; Digital–revised: 14 thousand USD; Print–retained: 34 thousand USD; Print–revised: 9 thousand USD

Payoff equation. $P_{ay}(A_i, S_j)$ = direct profit stated for the licence–syllabus pair. **Calculations.** No components need to be added: each symbolic payoff is assigned its stated numerical value.

Licence	Syllabus retained (S_1)	Syllabus revised (S_2)
Digital licence (A_1)	$P_{ay}(A_1, S_1) = 23$	$P_{ay}(A_1, S_2) = 14$
Print licence (A_2)	$P_{ay}(A_2, S_1) = 34$	$P_{ay}(A_2, S_2) = 9$

Final payoff table (profit in thousand USD).

Alternative	Syllabus retained (S_1)	Syllabus revised (S_2)
Digital licence (A_1)	23	14
Print licence (A_2)	34	9

1.2 Exercise 2: Rehearsal studio contract

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Daytime contract	A_2: Evening contract
States of nature	S_1: Extended festival	S_2: Shortened festival
Uncertain event	The festival duration announced by the independent organiser after the studio contract is chosen	
Consequences	Profit in thousand JPY for each alternative–state pair. Components: alternative-specific fixed amount, state-specific fixed adjustment.	



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Daytime contract; A_2: Evening contract	Alternative labels, in row order.
S_j	S_1: Extended festival; S_2: Shortened festival	State labels, in column order; revealed after the choice.
$f^A(A_i)$	A_1: 640 thousand JPY; A_2: 480 thousand JPY	Fixed net contribution from the chosen studio contract.
$f^S(S_j)$	S_1: 210 thousand JPY; S_2: -120 thousand JPY	Fixed access payment or cancellation deduction, applied once to either contract.

Payoff equation.

$$P_{ay}(A_i, S_j) = f^A(A_i) + f^S(S_j).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2
	$f^A(A_1)$	$f^A(A_1)$
A_1	$+ f^S(S_1)$	$+ f^S(S_2)$
	$= P_{ay}(A_1, S_1)$	$= P_{ay}(A_1, S_2)$
	$f^A(A_2)$	$f^A(A_2)$
A_2	$+ f^S(S_1)$	$+ f^S(S_2)$
	$= P_{ay}(A_2, S_1)$	$= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
	640	640
A_1	$+210$	-120
	$=850$	$=520$
	480	480
A_2	$+210$	-120
	$=690$	$=360$

Negative fixed amounts are deducted once. A negative per-unit adjustment reduces the contribution per completed unit; a negative payoff is a shortfall.

Final payoff table (profit in thousand JPY).

Alternative	Extended festival (S_1)	Shortened festival (S_2)
Daytime contract (A_1)	850	520
Evening contract (A_2)	690	360

1.3 Exercise 3: Archive imaging equipment

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1**:** Flatbed scanner	A_2**:** Overhead camera
States of nature	S_1**:** Dry air	S_2**:** Humid air
Uncertain event	Ambient humidity during the archive project, revealed after the imaging equipment is selected	
Consequences	Profit in thousand CNY for each alternative–state pair. Components: completed output quantity, alternative-specific net contribution per unit, state-specific fixed adjustment.	



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Flatbed scanner; A_2: Overhead camera	Alternative labels, in row order.
S_j	S_1: Dry air; S_2: Humid air	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	A_1: 13, 9 collections; A_2: 11, 7 collections (state order)	Completed archive collections for each equipment–humidity pair.
$v^A(A_i)$	A_1: 7 thousand CNY/collection; A_2: 9 thousand CNY/collection	Net contribution per completed collection from the chosen equipment.
$f^S(S_j)$	S_1: 8 thousand CNY; S_2: -7 thousand CNY	Fixed preservation bonus or climate-control loss for either choice.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v^A(A_i) + f^S(S_j).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2
	$q(A_1, S_1) \times v^A(A_1)$	$q(A_1, S_2) \times v^A(A_1)$
A_1	$+ f^S(S_1)$	$+ f^S(S_2)$
	$= P_{ay}(A_1, S_1)$	$= P_{ay}(A_1, S_2)$
	$q(A_2, S_1) \times v^A(A_2)$	$q(A_2, S_2) \times v^A(A_2)$
A_2	$+ f^S(S_1)$	$+ f^S(S_2)$
	$= P_{ay}(A_2, S_1)$	$= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
	13×7	9×7
A_1	$+8$	-7
	$= 99$	$= 56$
	11×9	7×9
A_2	$+8$	-7
	$= 107$	$= 56$

Negative fixed amounts are deducted once. A negative per-unit adjustment reduces the contribution per completed unit; a negative payoff is a shortfall.

Final payoff table (profit in thousand CNY).

Alternative	Dry air (S_1)	Humid air (S_2)
Flatbed scanner (A_1)	99	56
Overhead camera (A_2)	107	56

1.4 Exercise 4: Ceramics production line

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Tableware line	A_2: Tile line
States of nature	S_1: Brisk wholesale orders	S_2: Quiet wholesale orders
Uncertain event	The level of wholesale orders received after the ceramics production line is selected	
Consequences	Profit in thousand EUR for each alternative–state pair. Components: completed output quantity, alternative-specific net contribution per unit, state-specific net adjustment per unit, alternative-specific fixed amount.	



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Tableware line; A_2: Tile line	Alternative labels, in row order.
S_j	S_1: Brisk wholesale orders; S_2: Quiet wholesale orders	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	A_1: 18, 11 batches; A_2: 15, 8 batches (state order)	Completed firing batches for each line–order pair.
$v^A(A_i)$	A_1: 5 thousand EUR/batch; A_2: 7 thousand EUR/batch	Net contribution per firing batch from the chosen line.
$v^S(S_j)$	S_1: 4 thousand EUR/batch; S_2: 2 thousand EUR/batch	Additional net contribution per batch under either line.
$f^A(A_i)$	A_1: 9 thousand EUR; A_2: 13 thousand EUR	Positive fixed partner contribution from the chosen production line.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1)\times$	$q(A_1, S_2)\times$
	$[v^A(A_1)+v^S(S_1)]$	$[v^A(A_1)+v^S(S_2)]$
	$+f^A(A_1)$	$+f^A(A_1)$
	$=P_{ay}(A_1, S_1)$	$=P_{ay}(A_1, S_2)$
A_2	$q(A_2, S_1)\times$	$q(A_2, S_2)\times$
	$[v^A(A_2)+v^S(S_1)]$	$[v^A(A_2)+v^S(S_2)]$
	$+f^A(A_2)$	$+f^A(A_2)$
	$=P_{ay}(A_2, S_1)$	$=P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$18\times$	$11\times$
	$[5+4]$	$[5+2]$
	$+9$	$+9$
	$=171$	$=86$
A_2	$15\times$	$8\times$
	$[7+4]$	$[7+2]$
	$+13$	$+13$
	$=178$	$=85$

Final payoff table (profit in thousand EUR).

Alternative	Brisk wholesale orders (S_1)	Quiet wholesale orders (S_2)
Tableware line (A_1)	171	86
Tile line (A_2)	178	85

1.5 Exercise 5: Vocational training programme

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description
Alternatives	A_1: Woodworking programme A_2: Metalworking programme
States of nature	S_1: Increased funding S_2: Baseline funding S_3: Reduced funding
Uncertain event	The regional authority's training-funding level, announced after the programme is selected
Consequences	Profit in thousand GBP for each alternative–state pair. Components: completed output quantity, alternative-specific net contribution per unit, state-specific net adjustment per unit, alternative-specific fixed amount, state-specific fixed adjustment.



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Woodworking programme; A_2: Metalworking programme	Alternative labels, in row order.
S_j	S_1: Increased funding; S_2: Baseline funding; S_3: Reduced funding	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	A_1: 12, 8, 3 courses; A_2: 16, 11, 6 courses (state order)	Completed courses for each programme–funding pair.
$v^A(A_i)$	A_1: 6 thousand GBP/course; A_2: 7 thousand GBP/course	Net contribution per completed course from the chosen programme.
$v^S(S_j)$	S_1: 3 thousand GBP/course; S_2: 2 thousand GBP/course; S_3: -2 thousand GBP/course	Per-course funding addition or deduction, for either programme.
$f^A(A_i)$	A_1: 9 thousand GBP; A_2: 11 thousand GBP	Positive fixed partner payment for the chosen programme.
$f^S(S_j)$	S_1: 8 thousand GBP; S_2: 0 thousand GBP; S_3: -6 thousand GBP	Fixed administration grant, zero adjustment, or administration loss for either programme.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i) + f^S(S_j).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2	S_3
	$q(A_1,S_1)\times$	$q(A_1,S_2)\times$	$q(A_1,S_3)\times$
	$[v^A(A_1)+v^S(S_1)]$	$[v^A(A_1)+v^S(S_2)]$	$[v^A(A_1)+v^S(S_3)]$
A_1	$+f^A(A_1)$	$+f^A(A_1)$	$+f^A(A_1)$
	$+f^S(S_1)$	$+f^S(S_2)$	$+f^S(S_3)$
	$=P_{ay}(A_1,S_1)$	$=P_{ay}(A_1,S_2)$	$=P_{ay}(A_1,S_3)$
	$q(A_2,S_1)\times$	$q(A_2,S_2)\times$	$q(A_2,S_3)\times$
	$[v^A(A_2)+v^S(S_1)]$	$[v^A(A_2)+v^S(S_2)]$	$[v^A(A_2)+v^S(S_3)]$
A_2	$+f^A(A_2)$	$+f^A(A_2)$	$+f^A(A_2)$
	$+f^S(S_1)$	$+f^S(S_2)$	$+f^S(S_3)$
	$=P_{ay}(A_2,S_1)$	$=P_{ay}(A_2,S_2)$	$=P_{ay}(A_2,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
	$12\times$	$8\times$	$3\times$
	$[6+3]$	$[6+2]$	$[6-2]$
A_1	$+9$	$+9$	$+9$
	$+8$	$+0$	-6
	$=125$	$=73$	$=15$
	$16\times$	$11\times$	$6\times$
	$[7+3]$	$[7+2]$	$[7-2]$
A_2	$+11$	$+11$	$+11$
	$+8$	$+0$	-6
	$=179$	$=110$	$=35$

Negative fixed amounts are deducted once. A negative per-unit adjustment reduces the contribution per completed unit; a negative payoff is a shortfall.

Final payoff table (profit in thousand GBP).

Alternative	Increased funding (S_1)	Baseline funding (S_2)	Reduced funding (S_3)
Woodworking programme (A_1)	125	73	15
Metalworking programme (A_2)	179	110	35

1.6 Exercise 6: Road resurfacing equipment

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Portable mixer	A_2: Towable mixer	A_3: Batching plant
States of nature	S_1: Dry weather	S_2: Mixed weather	S_3: Persistent rain
Uncertain event	Weather during the resurfacing period, revealed after the equipment is committed		
Consequences	Profit in million COP for each alternative–state pair. Components: completed output quantity, alternative-specific net contribution per unit, state-specific net adjustment per unit, alternative-specific fixed amount, state-specific fixed adjustment.		



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Portable mixer; A_2: Towable mixer; A_3: Batching plant	Alternative labels, in row order.
S_j	S_1: Dry weather; S_2: Mixed weather; S_3: Persistent rain	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	A_1: 16, 11, 4 sections; A_2: 20, 14, 5 sections; A_3: 24, 19, 6 sections (state order)	Completed road sections for each equipment–weather pair.
$v^A(A_i)$	A_1: 7 million COP/section; A_2: 8 million COP/section; A_3: 9 million COP/section	Net contribution per completed road section from the chosen equipment.
$v^S(S_j)$	S_1: 3 million COP/section; S_2: 0 million COP/section; S_3: -3 million COP/section	Per-section addition, zero adjustment, or reduction under every choice.
$f^A(A_i)$	A_1: -25 million COP; A_2: -38 million COP; A_3: -52 million COP	Negative fixed equipment commitment, deducted once in every state.
$f^S(S_j)$	S_1: 11 million COP; S_2: 0 million COP; S_3: -14 million COP	Fixed completion bonus, zero adjustment, or disruption loss under every choice.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i) + f^S(S_j).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2	S_3
	$q(A_1,S_1)\times$	$q(A_1,S_2)\times$	$q(A_1,S_3)\times$
	$[v^A(A_1)+v^S(S_1)]$	$[v^A(A_1)+v^S(S_2)]$	$[v^A(A_1)+v^S(S_3)]$
A_1	$+f^A(A_1)$	$+f^A(A_1)$	$+f^A(A_1)$
	$+f^S(S_1)$	$+f^S(S_2)$	$+f^S(S_3)$
	$=P_{ay}(A_1,S_1)$	$=P_{ay}(A_1,S_2)$	$=P_{ay}(A_1,S_3)$
	$q(A_2,S_1)\times$	$q(A_2,S_2)\times$	$q(A_2,S_3)\times$
	$[v^A(A_2)+v^S(S_1)]$	$[v^A(A_2)+v^S(S_2)]$	$[v^A(A_2)+v^S(S_3)]$
A_2	$+f^A(A_2)$	$+f^A(A_2)$	$+f^A(A_2)$
	$+f^S(S_1)$	$+f^S(S_2)$	$+f^S(S_3)$
	$=P_{ay}(A_2,S_1)$	$=P_{ay}(A_2,S_2)$	$=P_{ay}(A_2,S_3)$
	$q(A_3,S_1)\times$	$q(A_3,S_2)\times$	$q(A_3,S_3)\times$
	$[v^A(A_3)+v^S(S_1)]$	$[v^A(A_3)+v^S(S_2)]$	$[v^A(A_3)+v^S(S_3)]$
A_3	$+f^A(A_3)$	$+f^A(A_3)$	$+f^A(A_3)$
	$+f^S(S_1)$	$+f^S(S_2)$	$+f^S(S_3)$
	$=P_{ay}(A_3,S_1)$	$=P_{ay}(A_3,S_2)$	$=P_{ay}(A_3,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
	$16\times$	$11\times$	$4\times$
	$[7+3]$	$[7+0]$	$[7-3]$
A_1	-25	-25	-25
	$+11$	$+0$	-14
	$=146$	$=52$	$=-23$
	$20\times$	$14\times$	$5\times$
	$[8+3]$	$[8+0]$	$[8-3]$
A_2	-38	-38	-38
	$+11$	$+0$	-14
	$=193$	$=74$	$=-27$
	$24\times$	$19\times$	$6\times$
	$[9+3]$	$[9+0]$	$[9-3]$
A_3	-52	-52	-52
	$+11$	$+0$	-14
	$=247$	$=119$	$=-30$

Negative fixed amounts are deducted once. A negative per-unit adjustment reduces the contribution per completed unit; a negative payoff is a shortfall.

Final payoff table (profit in million COP).

Alternative	Dry weather (S_1)	Mixed weather (S_2)	Persistent rain (S_3)
Portable mixer (A_1)	146	52	-23
Towable mixer (A_2)	193	74	-27
Batching plant (A_3)	247	119	-30